

Video Processing – HW1

Harris Corner Detection and Intro to Video Processing

Course 0512.4263 2025/2026

Section 2 / Part 2.1 – Harris Corner Detector: Open Questions

Q1. Is the Harris corner detector invariant to translation?

Answer: Yes.

Explanation. Every step of the detector (gradients, pointwise products, convolution with g , the response R , NMS, and threshold) is shift-invariant: it depends only on relative pixel positions. So shifting the image by (dx, dy) shifts the corner map by the same (dx, dy) , and the same physical points are detected.

Q2. Is the Harris corner detector invariant to rotation?

Answer: Yes.

Explanation. The Harris response depends only on the eigenvalues of

$$M = \begin{bmatrix} S_{xx} & S_{xy} \\ S_{xy} & S_{yy} \end{bmatrix},$$

since $R = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2$. Rotating the image transforms M into $R_\theta M R_\theta^T$, which has the same eigenvalues. So R is unchanged (just evaluated at the rotated pixel location), and the same points are detected as corners.

Q3. Is the Harris corner detector invariant to constant illumination ($I_{out} = a \cdot I_{in} + b$)?

Answer: No.

Explanation. The bias b cancels in the gradients, so it has no effect. The gain a scales the gradients by a , hence S_{xx}, S_{yy}, S_{xy} by a^2 and the response by $R' = a^4 R$. NMS picks the same local maxima, but the fixed threshold passes a different set of pixels, so the binary corner map changes. Invariant to b , not invariant to $a \Rightarrow$ not fully invariant.

Section 3 – Q3: Sobel Operator

Describe the Sobel operator and explain how it may be used to obtain the edge map of each frame of the video.

Definition.

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix}.$$

Gradient magnitude / direction.

$$|G| = \sqrt{G_x^2 + G_y^2}, \quad \theta = \text{atan2}(G_y, G_x).$$

By applying the sobel operator on a frame of the video, we get a 2-dimensional map of the gradient at each point, compute the magnitude $|G|$ and threshold. Pixels with $|G|$ above the threshold are marked as edges; the rest are background. Repeating this on every frame produces an edge-map video.